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Lightweight, Strong, and Transparent Wood Films Produced by Capillary Driven Self-Densification

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Wood delignification and densification enable the production of high strength and/or transparent wood materials with exceptional properties. However, processing needs to be more sustainable and besides the chemical delignification treatments, energy intense hot-pressing calls for alternative approaches. Here, this study shows that additional softening of delignified wood via a mild swelling process using an ionic liquid-water mixture enables the densification of tube-line wood cells into layer-by-layer sheet structures without hot-pressing. The natural capillary force induces self-densification in a simple drying process resulting in a transparent wood film. The as-prepared films with ≈150 µm thickness possess an optical transmittance ≈70%, while maintaining optical haze >95%. Due to the densely packed sheet structure with a large interfacial area, the reassembled wood film is fivefold stronger and stiffer than the delignified wood in fiber direction. Owing to a low density, the specific tensile strength and elastic modulus are as high as 282 MPa cm^3 g^{-1} and 31 GPa cm^3 g^{-1}. A facile and highly energy efficient wood nanotechnology approach are demonstrated toward more sustainable materials and processes by directly converting delignified wood into transparent wood omitting polymeric matrix infiltration or mechanical pressing.

1. Introduction

During billions of years of evolution, nature has developed a broad range of hierarchical materials. Nanostructured biological materials, such as wood,[1] silk,[2] chitin,[3] and well-known organic–inorganic composite materials, like nacre,[4] are formed through the assembly of repetitive building blocks into higher-order architectures.[5] This hierarchical design principle provides these materials with remarkable performances and thus makes them attractive as role models for the development of mechanically strong, eco-friendly and biocompatible materials for various applications.[5] Currently, bio-inspired multiscale assembly approaches are mainly based on a disassembly step to isolate the building blocks and a subsequent bottom-up assembly,[6–8] for instance, by 3D printing.[9] However, sustainable upscaling of nanostructures to the macroscale remains a challenge from an engineering perspective.

Wood is a renewable hierarchical material, consisting of aligned wood fibers with cell walls built by bottom-up assembly of high-strength cellulose microfibrils and matrix components (hemicelluloses and lignin).[1] The historical utilization of wood represents an impressive example of how humans have used a hierarchical biological material for their desired applications. About thousand years ago, Chinese started producing a paper with good mechanical

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The ORCID identification number(s) for the author(s) of this article can be found under https://doi.org/10.1002/smll.202311966

DOI: 10.1002/smll.202311966

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properties, known as Xuan paper, which was made up of mechanically disassembled multiscale cellulose fibers from plants including wood and rice straw.[10] Nowadays, various bottom-up fabrication strategies are explored based on using extracted cellulose nanofibrils (CNFs) and cellulose nanocrystals (CNC) from cell walls of wood or other plant resources.[11,12] By removing water, CNFs in water-based suspension can be reassembled into functional materials, such as transparent nanopaper,[6] strong microscale fibers.[13] or ultra-porous aerogels.[14] However, the isolation of CNFs requires large consumptions of energy and, most importantly, the beneficial fiber alignment and structural hierarchy of the starting material is inevitably lost.

In contrast, recently emerged wood nanotechnology employs a top-down approach that retains the anisotropic structure of wood and has attracted increasing attention for the engineering of functional wood materials, such as the production of transparent wood (TW).[15–18] In this top-down approach, lignin can be removed or decolorized by a structure retaining chemical treatment, resulting in delignified wood (DW).[19] or decolorized white wood,[20,21] followed by infiltration with a polymer precursor, forming a polymer with a refractive index matching that of cellulose ≈1.53,[22] like fossil-based poly(methyl methacrylate) (PMMA).[19] or bio-based poly(limonene acrylate) (PLIMA).[23] The obtained TW possesses attractive optical properties with both high optical transmittance and haze. However, a low interfacial strength between delignified cell walls and polymer matrix due to chemical incompatibility can be a limiting factor in terms of mechanical performance.[23,24] Furthermore, in case of common fossil-based resin systems, the matrix of these TWs diminishes their advantage as a sustainable material. Alternatively, a direct densification of delignified or partially delignified wood via high-pressure (hot) pressing, can physically interlock the cell walls, avoiding additional resin systems to produce strong transparent wood films (TWFs).[24,25] Nevertheless, this approach allows only for thin tissue sheets, and hot pressing is an energy intense process and the mechanically "fragile" delignified wood may crack especially perpendicular to fiber direction, thus limiting application at large scale.

Herein, inspired by the natural phenomenon of capillary action in porous media, we report on a capillary force driven self-densification by which the delignified wood cells collapse into layer-by-layer sheet structures, resulting in a TWF with substantially enhanced mechanical performance. The basic concept of this capillary self-densification process is schematically depicted in **Figure 1a**, showing aligned flexible high-aspect ratio fibers collapsing due to capillary forces during a drying process. Capillary force-induced aggregation.[26–28] has been widely used to manipulate several high-aspect ratio structures such as carbon nanotubes,[29] polymer micropillars.[30] and even hair.[31] In a previous study TEMPO-mediated oxidation was applied on delignified wood to soften the wood cell wall through in situ nanofibrillation, resulting in capillary force induced self-densification of wood during air-drying.[32] In our study we show that sufficient softening of the rigid cell walls can be obtained by a localized swelling by using ionic liquids (ILs), enabling capillary force driven reassembly of wood cell walls. ILs are well known as highly efficient and green solvents for lignocellulose swelling/dissolution and therefore, meet the requirements for realizing in situ swelling of delignified wood cell walls.[33,34] In recent studies, ILs induced swelling has been used as a pretreatment for the nanofibrillation of wood fibers.[35,36] By just diluting with water or common organic solvents, the solvent power of IL can be easily tuned from dissolving to swelling, the latter normally leading to significantly increased fiber diameters and cell wall thicknesses.[34,37] It has been previously reported that it is possible to generate high aspect ratio CNFs through swelling induced disintegration of thicker CNF bundles in water-containing IL with a water content of even up to 85 wt%.[6] Here we hypothesize, that the delignification step induces sufficient cell wall porosity.[38] for a subsequent infiltration with the swelling solvent (IL-water mixture), leading to an in situ swelling at the micro- or nano-scale to soften the cell wall. The capillary force aided self-densification is then obtained by a final drying step, after washing the swollen wood in water for solvent removal (Figure 1b). Applying this simple processing concept allows to obtain self-densified, transparent and strong wood films, while avoiding additional polymer infiltration and pressing (Figure 1c).

2. Results and Discussion

Low density balsa wood composed of fibers, parenchyma cells and vessels with thin cell walls was used in this work, likewise in former top-down wood nanotechnology studies.[15,21,24,39–44] First, we used an aqueous-based chemical delignification process (see details in the Experimental Section) to remove almost all lignin (see Table S1, Supporting Information) from the cell walls and thus to obtain white delignified wood (**Figure 2a**) as reported in previous studies.[45,46] The hollow wood cell structure was preserved after delignification and drying, and the cells were still connected at the tissue level (Figure 2b). Lignin removal results in a transverse cell wall softening, but not to an extent, which would allow for an intense self-densification during drying by capillary action. After air-drying at ambient condition, the obtained white DW cube samples showed only a reduction of ≈20% in thickness and ≈12% in width (Figure 2a), due to the apparent buckling of cell walls (Figure 2b). This indicates that wood cell walls after delignification still lack the necessary flexibility to allow for an efficient and extensive capillary force driven cohesion and collapse of cells.

To further soften the cell walls and thus to endow capillary collapse of cells, we adopted a physical swelling strategy using an ionic liquid (IL, 1-ethyl-3-methylimidazolium acetate, [EMIM][OAc])−20 wt% water mixture as non-derivative swelling agent. This mixture solvent is known to be unable to dissolve cellulose, but possesses strong swelling ability most likely owing to relatively larger β values (hydrogen-bond basicity, the ability to accept hydrogen bonds, correlating to the breakage of hydrogen bonds between cellulose) as compared to that of water.[34,37,47] As shown in our previous study swelling using an IL-water mixture enables further disintegration of thicker CNF bundles into thinner CNFs supported by cryo-TEM and rheology.[6] This swelling process allows for additional softening of the delignified wood cell walls. In current work, the incorporation of 20 wt% of water tuned the solvent power of IL from dissolution to swelling as above-mentioned, and strongly reduced the viscosity of IL (Figure S1, Supporting Information), which guaranteed an efficient and sufficient infiltration of dried DW with this IL-water mixture.

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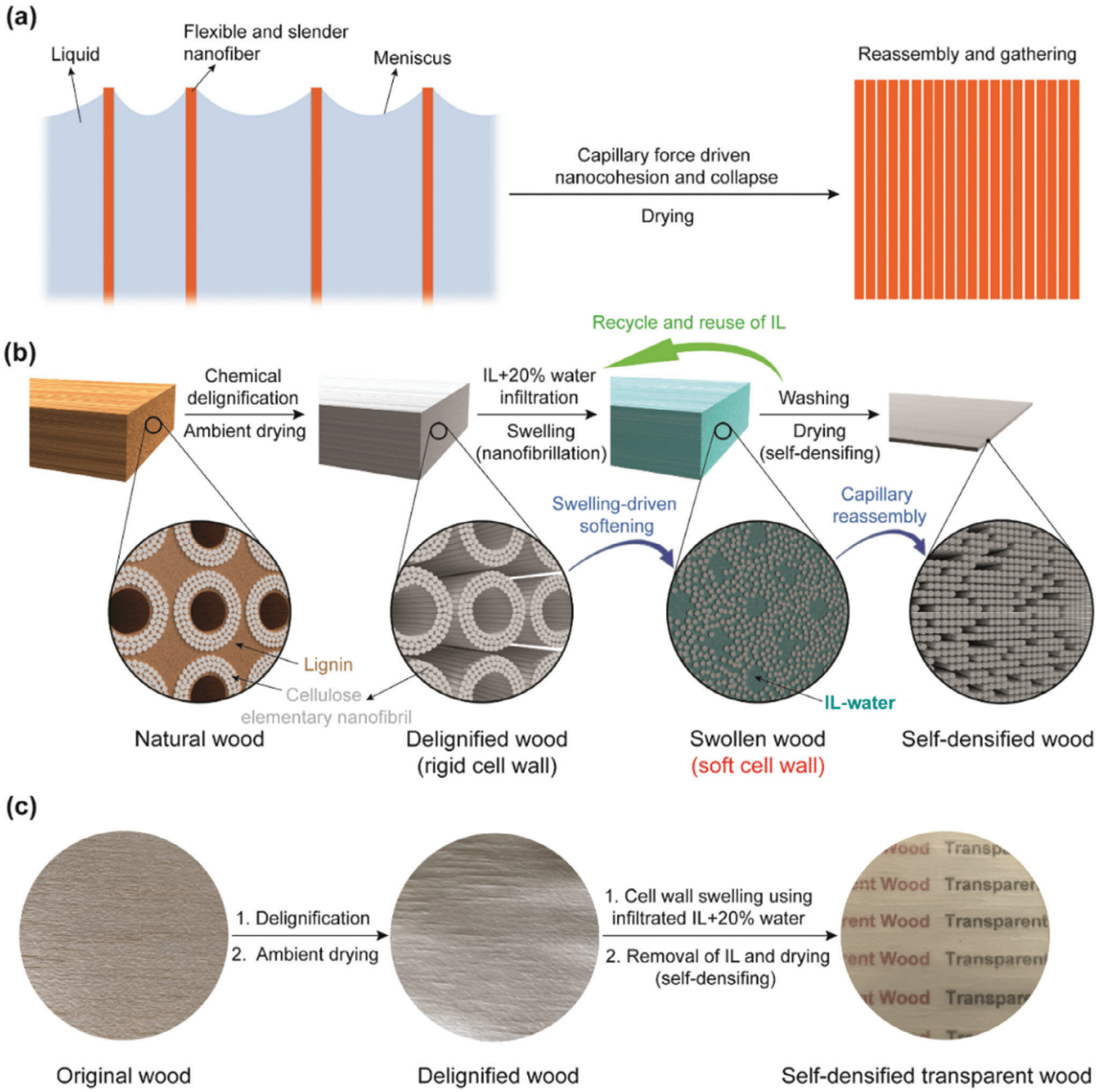

Figure 1. A top-down cell wall reassembly strategy through delignification, swelling induced cell wall softening and capillary force driven self-densification to produce self-densified transparent wood film (TWF). a) Schematic illustration of capillary force induced cohesion and gathering process of aligned flexible nanofibers upon drying. b) Schematic illustration of the wood cell wall structures (main part secondary layer) during the fabrication of self-densified wood, highlighting the critical role of the cell wall swelling process enabled by infiltration of IL-water mixture solvent, which softens the cell wall and thus facilitates the reassembling of wood cell walls into self-densified wood driven by capillary force. c) Photographic illustration of reassembling balsa wood veneer (thickness, ≈1 mm) to self-densified TWF (thickness, ≈0.15 mm) via a top-down approach, color changing from yellow original wood to white delignified wood and finally to a TWF.

We used optical microscopy to study the structural changes of wood fibers during the swelling process. In contrast to the observations on wet DW after mere delignification (Figure S2, Supporting Information), the cell walls of wood fibers were highly swollen with a core-shell structure owing to swelling after the 2 h vacuum-aided infiltration process at room temperature (Figure S3, Supporting Information). This is in accordance with previous observations on natural flax fibers swollen in NMMO (N-methylmorpholine N-oxide) containing more than 19–20 wt% of water and in [EMIM][OAc] containing 15 wt% water.[34,48] Moreover, the swollen DW in IL-water showed a much higher transparency than that in mere water (Figure S4,

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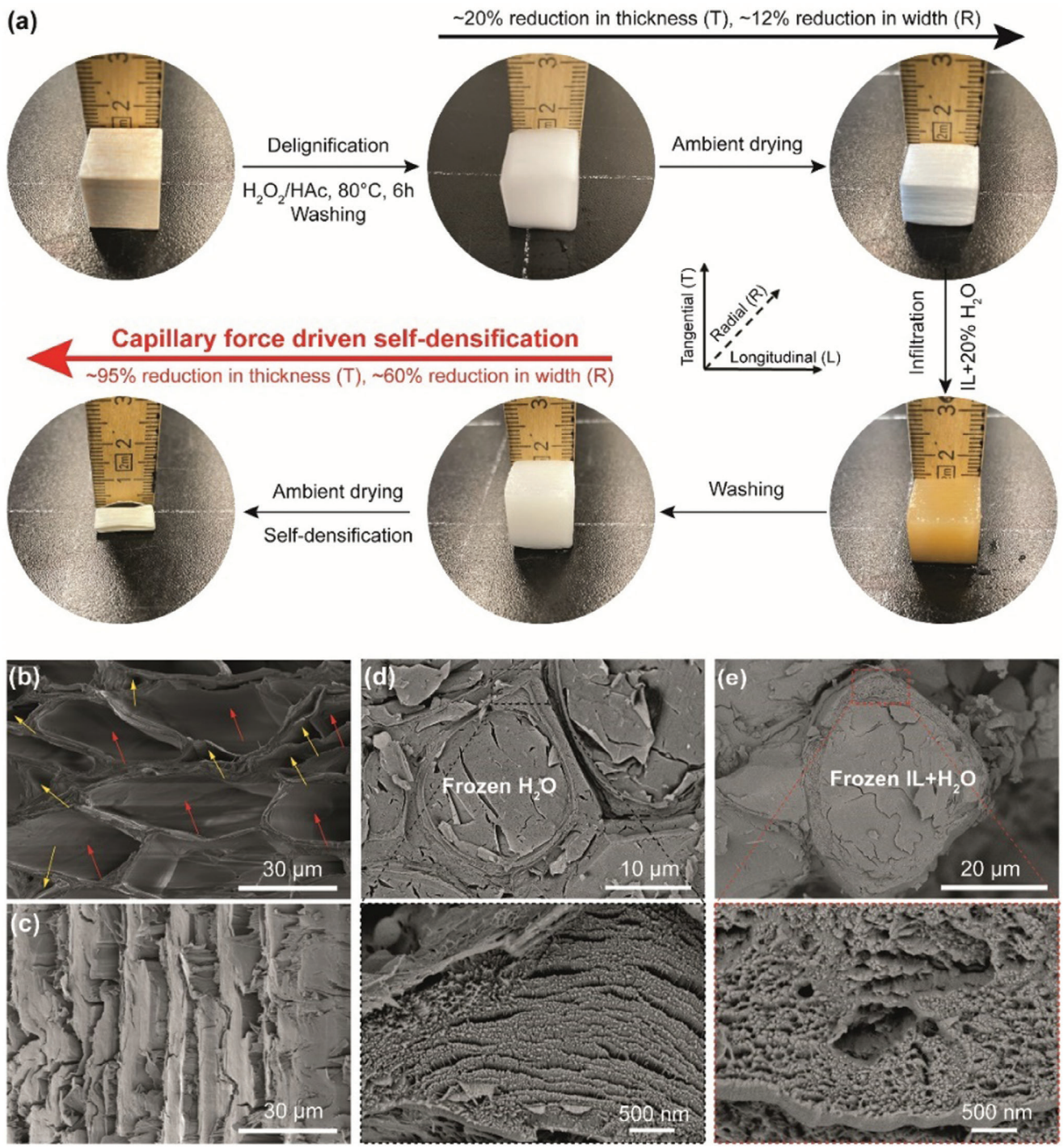

Figure 2. Capillary force induced self-densified wood preparation and microstructure characterization. a) Photograph illustration of the self-densification process of wood. b) SEM images of the cross-section of delignified wood, indicating efficient removal of lignin from cell wall corners (yellow arrow) and middle lamella (red arrow). c) SEM images of the cross-section of self-densified wood. d) Cryo-SEM imaging of the cell walls at wet state after delignification in water. e) Cryo-SEM imaging of the swollen cell wall in IL-20 wt% water mixture.

Supporting Information), which was also observed when in situ TEMPO oxidating DW.[32] Afterwards, the swollen samples were washed in deionized water to rinse out the IL and left for air-drying at ambient conditions. Because of the additional cell wall softening realized by the IL-water swelling strategy, the capillary forces during water evaporation in the air-drying process resulted in a severe structural collapse. We observed a substantial volume shrinkage of the DW cube in two directions (Figure 2a), specifically, ≈95% in thickness (tangential) and ≈60% in width (radial) direction, forming a self-densified and layered microstructure as

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visualized by SEM cross section images (Figure 2c) and side view image (Figure S5, Supporting Information). This two-directional densification enabled by capillary force driven cohesion is apparently different from mechanical pressing in one-direction to obtain a densified structure. The anisotropic size reduction is most likely influenced by gravity that promotes the capillary force driven aggregation in the direction of sample thickness. It is worth noting that we further probed the IL-water swelling induced structural change of the cell walls at nanoscale directly via cryo-SEM imaging, by which the cell wall structures in wet state were well maintained with the frozen water (Figure 2d) and IL-water (Figure 2e) filled in the lumen. Investigation at higher magnification (Figures 2d,e) indicates structural reorganization at the cell wall level by the IL-water swelling, potentially including an increase in cell wall porosity. Overall, these results allow to conclude that self-densification of delignified wood can be realized through cell wall swelling-softening followed by capillary force induced collapse during drying.

Optical transmittance scales with sample thickness, the lower the thickness the higher the transmittance.[49] Given the strong self-densifying effect on thicker wood cubes especially in the thickness direction, a self-densified transparent wood film can be produced from a thin balsa veneer applying the abovementioned treatment protocol. Similarly, a 1 mm-thick balsa wood veneer was delignified, air-dried, swollen over night after 2 h IL-20 wt% water mixture infiltration, washed and finally dried. To accelerate the drying process and prevent the warping of the sample during drying, the wet sample was loaded with ≈300 g weight on top during drying in a vacuum oven at 60 °C for 4 h (see details in the Experimental Section). For comparison, we also prepared control dried DW sheets without IL-20 wt% water swelling using the same drying procedure. The image presented in **Figure 3a** shows the control sample, which is opaque like the DW cubes (Figure 2a) and was named delignified wood sheet (DWS). The DWS featured apparent buckling of cell walls (Figure 3b), but due to limited capillary force driven cohesion, the cell lumina can still be clearly observed in a side view (Figure 3c). Additionally, the control indicates that the external weight applied during the drying process did not contribute much to cell collapse. In contrast, the IL-water treated wood veneer completely collapsed into a flexible transparent film (TWF) (Figure 3d; Figure S6, Supporting Information) with layer-by-layer sheet structure, as can be seen on images of the cross section (Figure 3e) and on the longitudinal side (Figure 3f). Delignified wood can also be collapsed into a transparent film with a laminated structure by external mechanical pressing.[24,50,51] However, this one-directional pressing folds the cell walls until cells come into close contact, and as a result, the cell wall contours remain visible. Additionally, as mentioned above IL-20 wt% water was selected in such way that no dissolution of cellulose occurs, we further compared the crystalline phase of the DWS and TWF by using X-ray diffraction (XRD), which both showed typical cellulose I allomorph and no obvious regenerated cellulose II allomorph (Figure 3g).

Small-Angle-X-ray Scattering (SAXS) was applied to provide a qualitative assessment of the change of the nanostructural organization of the cell walls and to gain further insight into the nanostructural effects of the IL-water swelling and the capillary driven self-densification. The representative SAXS curves (I(q) vs q) for the wet DW sheet before drying, dry DW sheet, swollen wood in IL-water, swollen wood after washing out IL and the as-prepared TWF (named as Wet DW, Dry DW, Swollen wood in IL-water, Swollen wood and TWF, respectively) are presented in Figure 3h, from which we further extracted the data as a Kratky plot ($I(q)*q^2$ vs q) in Figure 3i (more data is shown in the Figure S7, Supporting Information) to calculate the interfibrillar distances (see details in the experimental part). The wet DW showed an interfibrillar distance of 6.3 nm and decreased to 3.1 nm after ambient drying (Dry DW). Interestingly, swelling wood in the IL-water mixture (Swollen wood in IL-water) resulted in the same interfibrillar distance as that in water (Wet DW), this is surprising in view of the observed swelling behavior of the cell walls and the obvious impact of the IL-water swelling treatment on the nanostructure of cell walls (Figure 2d,e). The additional washing step further increased the distance to 8.4 nm (swollen wood), while the final drying step to obtain TWF led to the shortest interfibrillar distance of 2.9 nm. Note that although showing a comparable interfibrillar distance as that of the IL-water swollen wood, the cell walls of the mere water swollen wood (Wet DW) were still too rigid to collapse under the capillary forces. Therefore, in case of swollen wood in IL-water, besides in situ nanofibrillation, we suggest that the IL-water swelling treatment may also have a prominent impact on the cell wall at the ultrastructural level by a nonreversible softening, which accounts for the increase of the interfibrillar distance after washing out IL. However, further investigations are required.

To further investigate the impact of this capillary force driven cell wall densification on optical properties, we first compared the density and thickness values of DWS and TWF (Figure 3j) as they largely determine the transmittance and haze. After delignification and oven drying (DWS), the density slightly increased from ≈0.14 to ≈0.23 g cm^{-3} and the thickness decreased from ≈1 to ≈0.6 mm. In contrast, the self-densified TWF clearly demonstrated a substantially increased density of ≈0.8 g cm^{-3} with an apparently decreased thickness corresponding to ≈0.15 mm. As a result, the self-densified TWF showed a superior total optical transmittance of ≈70% compared to ≈20% of DWS at a wavelength of 550 nm (Figure 3k). The transmittance of the self-densified TWF is lower than that of reported TWF densified by external mechanical pressing, the latter reaching ≈80% at 550 nm wavelength, but these samples had a much lower thickness of≈55 μm.[24] Interestingly, the large increase in optical transmittance of the self-densified TWF compared to DWS is not accompanied by a strong decrease in haze (Figure 3l), since only a slight reduction was observed (DWS: ≈98%, TWF: 95%). We assume that the high haze of self-densified TWF might be due to a still low density for a densified sample and the "nano"-sheet structure (Figure 3e), leading to many cellulose sheet-air interfaces on the path of light through the TWF, which induce intense light scattering. The high transparency and high haze of TWF make it a promising green photonic material for optoelectronic applications such as solar cells, as haze increases light scattering and absorption in the active solar cell layer.[52,53]

Figure 4a compares the tensile mechanical properties by means of stress–strain curves of the self-densified TWF and control DWS in the fiber direction, which both show a linear deformation behavior until tensile failure. The mean values of tensile strength, elastic modulus and work of fracture of self-densified TWF are ≈550%, 500%, and 700% higher than the corresponding

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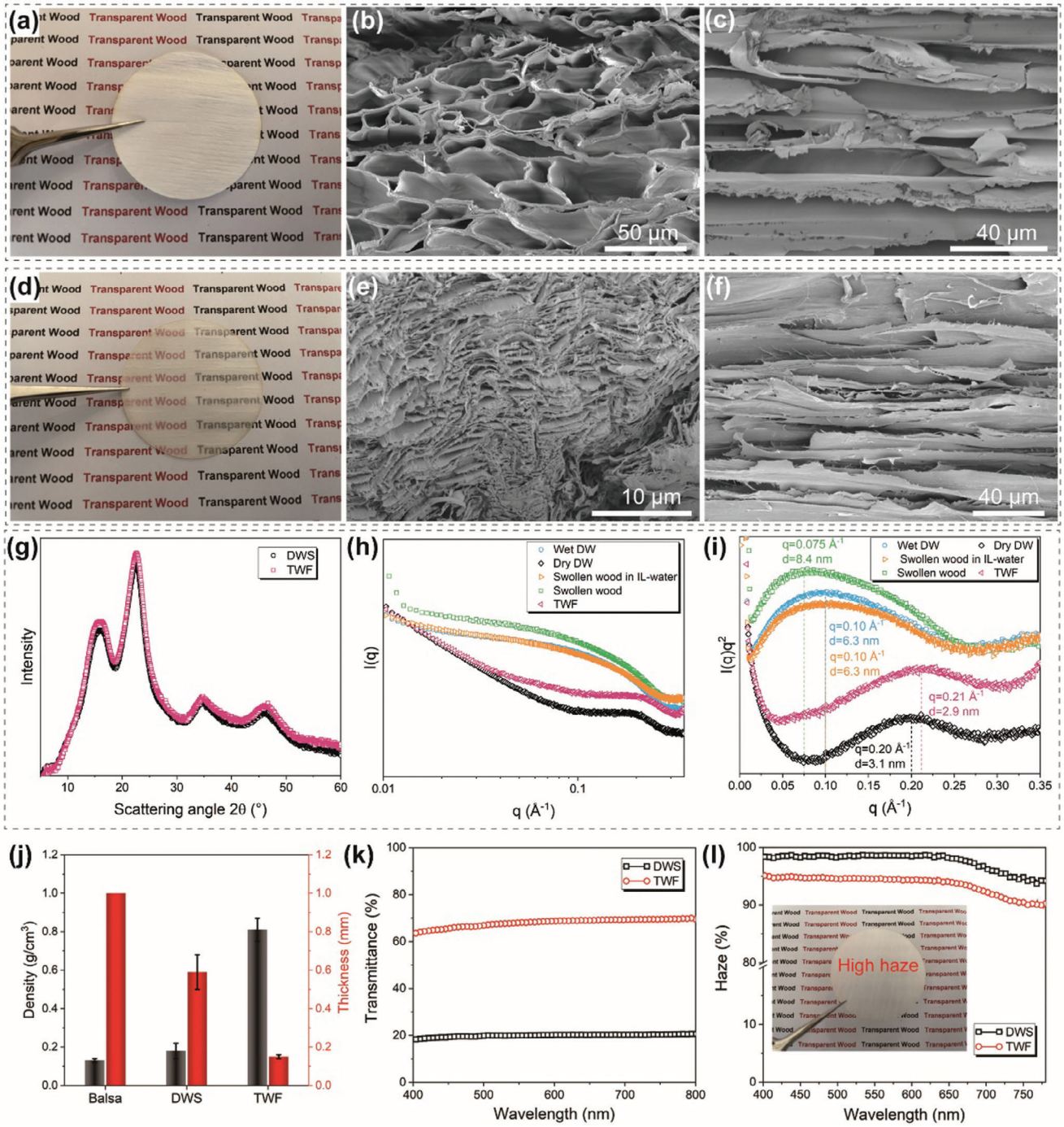

Figure 3. Morphological structures and optical properties of capillary force induced self-densified transparent wood film (TWF). a) Photograph of delignified wood sheet (DWS). b,c) SEM images of the freeze fractured cross-section and side view of DWS, respectively. d) Photograph of self-densified TWF. e,f) SEM images of the freeze fractured cross-section and side view of self-densified TWF. g) XRD diffractograms of DWS and TWF. h) SAXS double logarithmic plots of wet DW sheet, dry DW sheet, swollen wood in IL-water, swollen wood after washing out IL and the as-prepared TWF. i) The representative $I(q)q^2$ versus q profiles from SAXS for wet DW sheet, dry DW sheet, swollen wood in IL-water, swollen wood after washing out IL and the as-prepared TWF, see more curves in the Supporting Information Figure S7 (Supporting Information). The dashed lines in the plot indicate the maximum intensity peaks from which the interfibrillar distances are determined. j) Density and thickness of initial balsa, DWS and TWF. k) Optical transmittance of DWS and self-densified TWF. l) Optical haze of DWS and self-densified TWF, the digital image illustrates the high haze of TWF, with the sample being 1 cm away from the same substrate.

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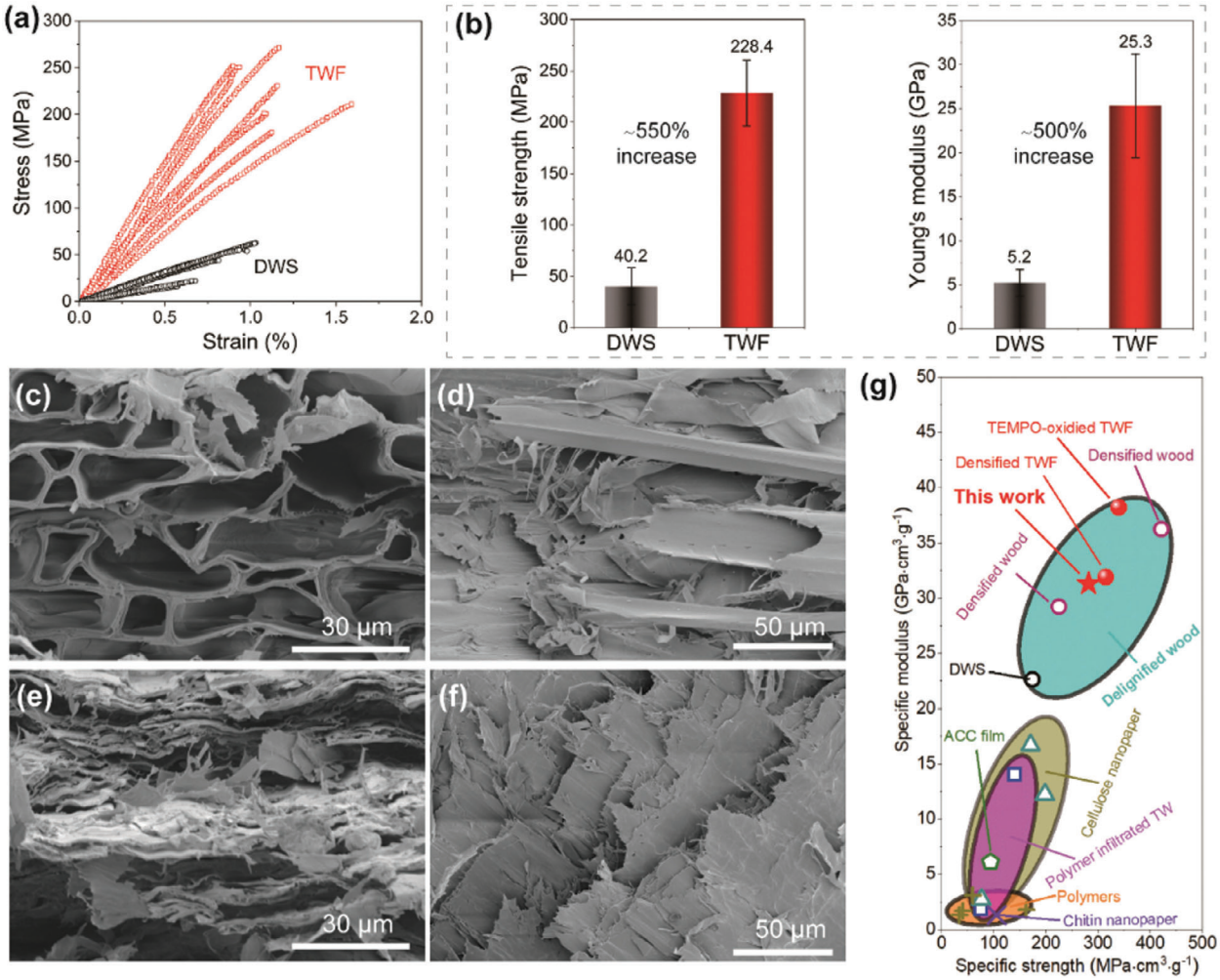

Figure 4. Mechanical properties and fracture morphologies of capillary force induced self-densified TWF. a) Typical stress–strain curves of self-densified TWF and DWS in the fiber direction. b) Comparison of the mechanical characteristics of control DWS (black bars) and self-densified TWF (red bars). c,d) SEM images of the fractured cross-section and fractured surface of DWS, respectively. e,f) SEM images of the fractured cross-section and fractured surface of self-densified TWF. g) The Ashby diagram of specific Young's modulus versus specific strength for various delignified wood derived materials,[19,23–25,50,51,54] cellulose nanopaper,[6,56,57] all-cellulose composite (ACC) film,[52] chitin nanopaper.[58] and polymer films.[59,60] Details of the mechanical properties presented in the Ashby plot are included in Table S2 (Supporting Information).

values of DWS (Figure 4b; Figure S8, Supporting Information). It is worth noting that the air dried delignified wood sheet showed much lower mechanical properties, with a tensile strength of 20.1 ± 3.4 MPa and an elastic modulus of 3.7 ± 0.7 GPa (Figure S9, Supporting Information). For a comparison between the tensile fracture morphologies of the control DWS and of the self-densified TWF, which may provide insights into the strengthening mechanisms of the TWF, we conducted SEM imaging of tensile fracture surfaces. Tensile failure of DWS clearly arises from tearing of microscale single cell wall fibers (Figure 4c,d). Contrary, the failure of the self-densified TWF initiates from tearing of the layer-by layer stacked cellulose "nano"-sheets along the fracture surface (Figure 4e,f). These observations indicate that the strengthening mechanisms of self-densified TWF originate from the layer-by layer stacked cellulose sheets and the closer packing of the cellulose fibrils within the sheet. These result in a strongly increased interfacial connection between aligned cellulose fibrils for an efficient stress transfer, presumably accompanied by with an increased hydrogen bonding density, thus consequently making a dominant contribution to the remarkably enhanced strength and Young's modulus.[54,55]

We further investigated the tensile behavior in the transverse direction to better understand the impact of the treatment on sample anisotropy. In consistency with previous studies, delignified wood and DWS were mechanically weak in the transverse direction with a tensile strength less than 2 MPa and a low elastic modulus of less than 0.2 GPa (Figure S10, Supporting Information). Tensile failure of the highly anisotropic DWS in the transverse direction originated from separation of the aligned cell fibers, resulting in a smooth fracture surface (Figure S11, Supporting Information). Interestingly, the self-densified TWF exhibited an average tensile strength of 20.9 ± 1.1 MPa and an elastic modulus of 3.0 ± 0.1 GPa in the transverse direction, which are ≈10 times and ≈15 times higher than the

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corresponding value of control DWS, respectively (see the representative stress–strain curves in Figure S12, Supporting Information). This substantially higher values correspond to a much lower mechanical anisotropy of TWF samples. Again, this can be explained by the increased interfacial connection between cellulose fibrils between the densely packed cellulose sheets (Figure S13, Supporting Information).

Generally, density has a major influence on the mechanical properties of cellulosic materials.[52,61] Therefore, similar and even superior mechanical performances were reported previously for mechanically pressure-densified materials from (partially) delignified wood,[24,50,51] since these densified wood samples usually have a higher density (> 1.2 g cm^{-3}) as compared to self-densified TWF (0.8 g cm^{-3}) (details of the mechanical properties and corresponding density values can be found in Table S1, Supporting Information). Taking the low density into consideration, our self-densified TWFs demonstrate a high specific tensile strength and an elastic modulus of 281.7 ± 39.5 MPa cm^3 g^{-1} and 31.3 ± 7.3 GPa cm^3 g^{-1}, respectively, as can be seen in the diagram in "Ashby-plot format" in Figure 4g and is further detailed with regard to mechanical properties obtained in former studies in Table S2 (Supporting Information). Additionally, we believe that both mechanical performance and optical transmittance of our lightweight TWF could be further enhanced by optimized fabrication processing as there is still room to increase the density to more than 1.2 g cm^{-3}. For instance, after the washing step (removal of IL), one could use instead of water an alternative solvent possessing a higher surface tension, which may effectively enhance the capillary force upon drying and therefore promote self-densification.

In this study, cell wall softening was obtained by physical IL-water swelling, which is beneficial in terms of sustainability and biodegradability. ILs are still expensive mainly due to recycling costs, in particular when very low water contents must be reached or solubilized material needs to be removed. However, in this study, we used IL-water mixture as a mere swelling agent at a water content of more than 20 wt%, which facilitates a potential recycling and can be easily reused. For instance, through vacuum and heating-induced water evaporation, we can efficiently obtain an IL-water mixture with a water content of 20 wt% from the diluted IL-water, which had been dissolved out of the modified wood, see Figure S14 (Supporting Information). Moreover, ILs have been demonstrated as promising carriers of organic/inorganic, nanoparticles and polymers for material modification and functionalization.[62–64] This may allow to simultaneously combine densification and functionalization of wood in a single step to address various advanced applications, in which lightweight and high load-bearing capacity are required, for example, green packaging and portable flexible electronics.

3. Conclusion

In conclusion, we realized a direct conversion of renewable wood into a lightweight and high-performance transparent film by a simple, yet effective capillary force driven self-densification approach enabled by physical swelling induced softening of delignified cell walls using solvent power tuned IL-water mixture. The porous tissue of wood was converted into a densely packed cellulose sheet structure due to capillary force driven structural reorganization at the cell wall level. The resulting self-densified TWF possesses a comparably low density but exhibits high mechanical properties. We attribute the enhanced mechanical performance in both fiber and transverse direction to the increased interfacial area in the self-densified TWF. The production of this fully wood-based transparent film could be scaled-up without the need of mechanical pressing. Moreover, different wood species and additional physical swelling treatments could be explored to manufacture self-densified TWF in the future. Overall, the studied fabrication strategy offers broad versatility in designing and manufacturing mechanically strong and functional wood-based materials for advanced applications.

4. Experimental Section

Materials: Balsa wood (Ochroma pyramidale) with an oven-dried density of ≈0.14 g cm^{-3} was purchased from a local DIY store and cut into cubes and veneers with a size of 15 × 15 × 15 mm^3 and 120 × 100 × 1 mm^3 (longitudinal × radial ×tangential), respectively. Glacial acetic acid (≥99.7 wt%) and hydrogen peroxide (30 wt%) were purchased from Sigma–Aldrich, Switzerland. The ionic liquid (IL) 1-ethyl-3-methylimidazolium acetate ([EMIM][OAc], > 95 wt%) was purchased from IoLiTec, Germany. The moisture content of [EMIM][OAc] was 0.27 wt% as determined in a previous study.[34] All chemicals were used as received and the water used throughout the study was deionized water.

Delignification: The wood cubes and radial cut veneers were delignified mainly following the protocol of previous studies,[50,65,66] but only with one pass herein. In short, the wood samples were delignified using an equal-volume mixture of acetic acid and hydrogen peroxide at 80 °C for up to 6 h. Then, the delignified samples were washed in water until the pH value of the water was neutral. After washing, the delignified wood (DW) cubes and sheets were dried under ambient conditions and stored in a climate room (20 °C, 65% relative humidity).

Self-Densification Process: The dried DW cubes/sheets were placed between two metal-grid sample holders and immersed with a pre-prepared mixture solvent (80 wt% IL + 20 wt% H_2O) in a beaker for infiltration under vacuum at room temperature for 2 h. To realize a sufficient swelling of the delignified cell walls, the beaker was sealed and kept at ambient condition overnight. To remove the IL from the swollen DW sample, it was kept in water at room temperature for ≈72 h (exchange with fresh water four times every day). The purified wet DW cubes were dried at ambient conditions to obtain a self-densified wood material. The wet DW sheets were covered by filter papers and a weight of 300 g to prevent warping, while being dried under vacuum in an oven at 60 °C for 4 h to obtain a self-densified transparent wood film (TWF). Additionally, wet DW sheets (without the IL treatment) were exposed to the same oven drying protocol to be used as a reference (named DWS).

Characterizations—Chemical Composition Analysis: The chemical composition (carbohydrates, lignin) of initial balsa, DWS and TWF was determined using quantitative saccharification method upon acid hydrolysis, according to NREL/TP-510–42618 norm. Monosaccharides (glucose, xylose, and mannose) were analyzed using a high-performance anion-exchange chromatography system that was equipped with pulse amperometric detection (HPAEC-PAD) and a Dionex ICS-3000 system, which were used to calculate the cellulose and hemicellulose contents. The acid-soluble lignin was determined by using a Shimadzu UV 2550 spectrophotometer at a wavelength of 205 nm using absorption coefficient of 110 L g^{-1} cm^{-1}.

Characterizations—Shear Viscosity: The shear viscosity of the IL and IL-water mixtures was characterized using a rheometer (Anton Paar MCR 302) equipped with a plate-plate geometry (25 mm plate diameter, 1 mm gap). Viscosity-shear rate dependencies were recorded from 1 to 100 s^{-1} at 25 °C.

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Characterizations—Optical Microscopy: The cell wall swelling of DW was studied by using a polarized optical microscopy (Olympus BX41-LED). A thin piece of swollen wood after 2 h IL-water infiltration was placed between two glass plates and was observed at room temperature. For comparison, a thin piece of wet DW was observed in the same way.

Characterizations—X-ray Diffraction (XRD): The XRD data of DWS and TWF in their sheet state were studied at room temperature by XRD (Panalytical X'Pert PRO MPD) equipped with a Ge monochromator using copper Kα1 radiation (λ = 1.5406 Å). The diffraction data were obtained using Bragg-Brentano geometry with an angular step size of 0.03° and a counting time of 1 s per step and recorded in the 2θ 5°–60° range.

Characterizations—Small-Angle-X-ray Scattering (SAXS): For the determination of the nanofiber distance, also referred to as interfibrillar correlation length, SAXS analysis was performed. Scattering patters were collected for delignified and IL-water treated specimens in the dry (dry DW sheet and TWF) and wet (wet DW sheet, swollen wood in IL-water and swollen wood after washing out IL) state. Dried samples were placed on a solid sample holder (Xenocs), whereas wet samples were taken from water/IL-water placed in a gel holder (Xenocs, the gel holder seals the wet sample between two Kapton foils, keeping the sample from drying during the experiment). The samples were placed in the sample holder with the fiber direction perpendicular to the incoming X-ray beam. The respective sample holder was placed in a laboratory SAXS system (Xenocs Xeuss 3.0), equipped with a Cu Kα X-ray source, and the 2D scattering signal was recorded on a Dectris EIGER2 1 M detector placed at a sample detector distance of 400 mm. The 2D signal was azimuthally integrated along the equatorial plane with an opening angle of 40° to obtain a 1D scattering signal. For the wet samples the background scattering of the empty geld holder was subtracted. All data handling was performed using the Xenocs XSACT software.[67] The nanofiber distance was estimated using the approach by Jakob and coworkers.[68] The dry scattering patterns have a distinct peak ≈2 nm^{-1} and the wet ones have a shoulder at ≈1 nm^{-1}. To be able to clearly determine the peak position, a proposed method was used.[69,70] of plotting the data as a Kratky plot (where I(q)*q^2 is plotted over q), which emphasizes the peaks and therefore helps determining the peak position q_{peak}. The nanofiber distance d is then calculated using $d = 2\pi/q_{peak}$.

Characterizations—Density Calculation: The density of DWS and TWF (after conditioned at 20 °C and 65% relative humidity) were determined by measuring weight and dividing by calculated volume.

Characterizations—Scanning Electron Microscopy (SEM): Freeze-fractured cross sections and tensile-fracture surfaces of the samples were observed by using a scanning electron microscopy (SEM) system (Hitachi SU5000) at an accelerating voltage of 2 kV. Prior to examination, the surfaces were coated with a 6 nm Pt–Pd (80/20) by a sputter-coater (CCU-010, Safematic).

Characterizations—Cryogenic Scanning Electron Microscopy (Cryo-SEM): The wet DW sheet and the IL-water containing swollen wood (before washing) were shock-vitrified by fast immersion into liquid nitrogen mud at 2100 atm in a self-designed cooling device, followed by cryo-fracture at −100 °C. The fractured specimens were then cryo-transferred to a special sputter-coater (Quorum PP 3010T). Prior to surface coating with a thin layer of Pt, the vitrified water on the surface of the specimens was sublimated under vacuum at −90 °C. Finally, the Pt coated specimens were cryo-transferred to the Cryo-SEM (Hitachi, Regulus 8100) operated at 4 V and −160 °C.

Characterizations—Ultraviolet–Visible Light (UV–vis) Spectroscopy: The transmittance and haze of DWS and TWF in the visible light region were obtained using a Lambda UV/Vis-650 with an Integrating Sphere Attachment (PerkinElmer, USA). Optical haze was used to quantify the percentage of light scattering and was calculated as follows:[52]

$$Haze\ (\%) = 100 \times \left(\frac{T_4}{T_2} - \frac{T_3}{T_1}\right) \quad (1)$$

where T_1, T_2, T_3, and T_4 are defined as background checking, total transmitted illumination, beam checking and pure diffusive transmittance, respectively.

Characterizations—Tensile Testing: The mechanical properties were studied by using a universal testing machine (Zwick, Germany) equipped with a 1 kN load cell and a video extensometer, the tests were conducted parallel to the grain. At least seven specimens with obtained thicknesses (DWS: ≈0.6 mm, TWF: ≈0.15 mm), 50 mm length × 5 mm width produced by laser cutting were conditioned for 24 h in the climate room (20 °C, 65% relative humidity). Both ends of the specimens were stabilized by attaching wood pieces (15 × 5 × 3 mm^3), to avoid slippage and reduce stress concentrations. For tensile tests, the sample were strained at a rate of 0.5 mm min^{-1}. In addition, the work of fracture was calculated by determining the area under the stress-strain curve.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work was financed in the framework of the project "Strong Composite", supported under the umbrella of ERANET Cofund ForestValue by Innosuisse, AKA, Business Finland, Vinnova, BMLFUW. ForestValue received funding from the European Union's Horizon 2020 research and innovation program under grant agreement 773324. F.C. gratefully acknowledges the financial support in the framework of the Excellent Discipline Cultivation Project by JHUN (2023XKZ034) and the National Key Research and Development Program of China (2022YFB3603702). Y.D. thanks the financial support from the SNF project "Hierarchical cellulose scaffolds for structural and functional gradient materials" (200021_184821/1). The authors thank Thomas Weber from the ETH Zürich X-ray Service Platform for the technical support with the SAXS equipment. Additionally, the authors thank Roman Elsener (WoodTec Group, Cellulose & Wood Materials, Empa) for the assistance with tensile testing measurements.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

capillary force, self-densification, swelling, transparent film, wood cell walls

Received: December 20, 2023
Revised: April 16, 2024
Published online: May 21, 2024

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